

Homework #3 – Financial Transactions

Star Schema and Interactive Dashboard — Summary Report

Big Data – Data Science course | Cristian Catania – ID 1000034809

1. Modeling Choices

The data warehouse was modeled as a star schema whose fact table records financial transactions. The grain is explicit: one row of Fact_Transactions represents one transaction from the account statement file. The grain is defined at the row level rather than per IDTransaction, because IDTransaction is not unique in the source (only 1,136 distinct values across 2,281 valid rows); the same identifier appears on different, unrelated transactions. A generated surrogate key (transaction_sk) is therefore used as the primary key, while IDTransaction is kept inside the fact table as a degenerate dimension.

The schema is composed of one fact table and four dimensions, summarized below.

Table	Type	Key columns / measure
Fact_Transactions	Fact	transaction_sk (PK); time_id, symbol_id, geo_id, type_id (FK); units (measure); IDTransaction (degenerate)
Dim_Time	Dimension	time_id (SK); date, day_of_week, month, quarter, year
Dim_Symbol	Dimension	symbol_id (SK); symbol, company_name, sector, industry
Dim_Geography	Dimension	geo_id (SK); country, alpha_2, sub_region, region
Dim_TransactionType	Dimension	type_id (SK); transaction_type

Dimension hierarchies

- Dim_Time: Day → Month → Quarter → Year. The attribute day_of_week is kept outside the hierarchy because it is required by a weekday-based query.
- Dim_Geography: Country → Sub-region → Region. Note: the assignment lists Country → Region → Sub-region, but the data shows that each region contains several distinct sub-regions (e.g. Europe contains Northern, Southern, Eastern and Western Europe) while each sub-region belongs to exactly one region. Country → Sub-region → Region was therefore adopted as the correct drill-up direction.
- Dim_Symbol: Symbol → Industry → Sector.
- Dim_TransactionType: a flat dimension with no hierarchy (BUY / SELL / DIVIDEND).

Design decisions

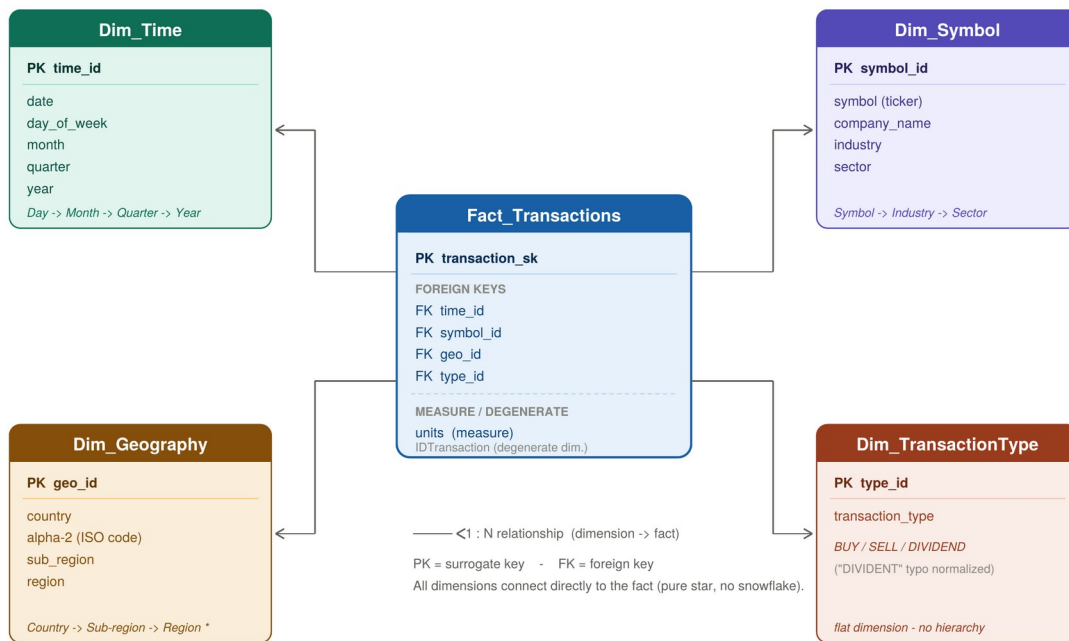
- Surrogate keys: each dimension uses an integer surrogate key, decoupled from the source natural keys.
- Pure star (no snowflake): although geography is derived through the symbol's country, geo_id is materialized directly in the fact table so that all four dimensions connect directly to the fact.

- Measure vs. derived: the only stored measure is units; transaction counts are computed at query time with COUNT and are not materialized.
- Unused attributes removed: from country.csv only name, alpha-2, sub-region and region were kept; the other columns (alpha-3, country-code, iso_3166-2, intermediate-region and the numeric *-code columns) were dropped as they are not used by any query.

Figure 1 — Star schema diagram.

Star Schema - Financial Transactions

Grain: one row of Fact_Transactions = one transaction from the account statement



* Dim_Geography hierarchy: the assignment states Country -> Region -> Sub-region. Country -> Sub-region -> Region was adopted instead because the data shows that each region contains several distinct sub-regions (e.g. Europe -> Northern/Southern/Eastern/Western Europe), whereas each sub-region belongs to exactly one region: the sub-region is therefore the intermediate level between country and region.

2. Data Quality Issues Found

Several data quality problems were detected during the basic checks required by the assignment (missing values, unused attributes, referential integrity of symbols and countries). Each issue and the adopted resolution is reported below.

Issue	Resolution
464 fully empty rows and a phantom column (Unnamed: 5) caused by the trailing ';' on every line.	Both removed; 2,281 valid transactions remain.
Transaction type typo: "DIVIDENT".	Normalized to "DIVIDEND".
IDTransaction is not unique (1,136 distinct values)	Not used as a key; surrogate transaction_sk used

Issue	Resolution
over 2,281 rows).	instead, IDTransaction kept as a degenerate dimension.
18 traded symbols (212 transactions, 9.3%) are absent from symbols.csv, so they cannot be enriched with sector / industry / country.	Transactions kept (totals stay correct) with an "Unknown" member; excluded only from rankings by enrichment attributes.
Country names "Turkey" and "Taiwan" do not match the official ISO names in country.csv.	Mapped explicitly to "Türkiye" and "Taiwan, Province of China".
country.csv has NULL region/sub-region for Antarctica and Taiwan.	Antarctica is never traded (irrelevant); Taiwan filled manually as Eastern Asia / Asia.
Namibia's ISO code "NA" was read as a missing value by pandas.	Prevented with keep_default_na=False, na_values=[""] when reading country.csv.

On the "Unknown" member: the 18 orphan tickers are real, well-known companies (e.g. SAP, MONC, UCG) that simply have no entry in the symbols reference. Grouping them under a single "Unknown" member is a simplification that slightly under-counts the affected countries in geographic rankings. They were not mapped manually because that would require external knowledge not contained in the provided datasets. The trade-off is documented and considered acceptable for this exercise.

3. Main Analytical Results

Five of the proposed questions were answered, chosen so that together they exercise every dimension. DIVIDEND transactions were excluded from all queries, since the questions concern BUY/SELL trading activity, and the "Unknown" member was excluded from rankings by enrichment attributes.

Q1 – Top 5 sectors by SELL transactions in the US (2024)

Sector	SELL transactions
Technology	158
Communication Services	58
Financial Services	55
Healthcare	50
Consumer Cyclical	48

Q3 – Quarters of 2024 ranked by total transactions (BUY + SELL)

Quarter	Total transactions
Q1	1076
Q2	580
Q3	260
Q4	255

Note: all four quarters of 2024 are present in the data; the decline reflects a real drop in trading activity over the year, with Q1 being by far the most active.

Q7 – Top 10 symbols by transactions (BUY + SELL, 2024)

Symbol	Total transactions
ARM	100
AMD	97
TSM	77
TIMB	76
FNC	66
GOOG	48
AMZN	47
MSFT	45
ARDX	43
BRFS	42

Q9 – Top 5 industries by units traded in Asia (2024)

Industry	Units traded
Semiconductors	5401
Auto Manufacturers	4769
Diagnostics & Research	1945
Internet Content & Information	1805
Aerospace & Defense	1799

Q12 – Top 3 sectors by units sold on Mondays (2024)

Sector	Units sold
Technology	4361
Healthcare	2108
Consumer Cyclical	1218

Across the analyses, the Technology sector and the semiconductor industry are consistently dominant, both by number of transactions and by traded volume, with companies such as ARM, AMD and TSM leading the most-traded symbols.

Deliverables

- Jupyter Notebook with the ETL logic, star schema creation and analytical queries.

- Streamlit application (zipped) implementing Page 1 – Time Analysis.
- Star schema diagram (Figure 1).
- This summary report.